MIL-I-27623C
10 January 1972
Superseding
MIL-I-27623B(ASG)

MILITARY SPECIFICATION

INDICATOR, ATTITUDE, REMOTE STANDBY, 2 INCH

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for 2-inch remote standby attitude indicators.

1.2 Classification. The indicators shall be classified as follows (see 6.2):

MS27150-1	Red and white integral lighting with bank angle scale at bottom
MS27150-2	Red integral lighting with bank angle scale at top
MS27150-3	Red and white integral lighting with bank angle scale at bottom and with improved antireflection characteristics
MS27150-4	Red integral lighting with bank angle scale at top with improved antireflection characteristics
MS27150-5	Red integral lighting with bank angle scale at top, improved antireflection characteristics, and pin G grounded

* 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-365	Silver Plating, Electrodeposited, General Requirements for
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practice
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MS27150	Indicator, Attitude, Remote, Standby
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

3.2 Components. The indicator shall consist of servo-driven mechanisms with amplifiers for actuating the pitch and roll gimbals, an indicator display consisting basically of a fixed miniature airplane, bank angle dial, bank angle index, and a moving two-colored spheroid background with a dashed horizontal line dividing the two colors, a pitch trim knob mechanism, external zeroing adjustments for pitch and roll, and a power-failure indicating device and associated wiring.

3.2.1 Components other than display elements that are visible from the front of the indicator under day or night lighting conditions shall be so painted or color coded that they will not distract from the display.

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall conform to applicable specifications as specified herein. When materials are used that are not specifically designated, they shall be entirely suitable for the purpose. The use of lightweight materials and weight-saving designs shall be a major consideration and their use shall be investigated and exploited to the greatest possible extent.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.4.3 Corrosive fumes. Materials, as installed in the indicator and under the service conditions specified herein, shall not liberate deleterious fumes.

3.4.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:31Z
Check the source to verify that this is the current version before use.

* 3.4.4.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing, intimate contact of dissimilar metals shall be avoided. Dissimilar metals are defined in MIL-STD-889.

3.4.5 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The indicator shall be so designed and contructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. The indicator shall be of such design that it may be installed in a cutout as shown on figure 1.

3.5.1 Maintainability. The mean corrective maintenance downtime (M_{ct}) shall be 100 minutes including the time required to unseal the indicator and hermetically seal it after repair. The maximum corrective maintenance downtime ($M_{max_{ct}}$) shall be not more than 300 minutes including the time required to unseal the indicator and hermetically seal it after repair.

* 3.5.2 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF), as defined in MIL-STD-781, of 750 hours.

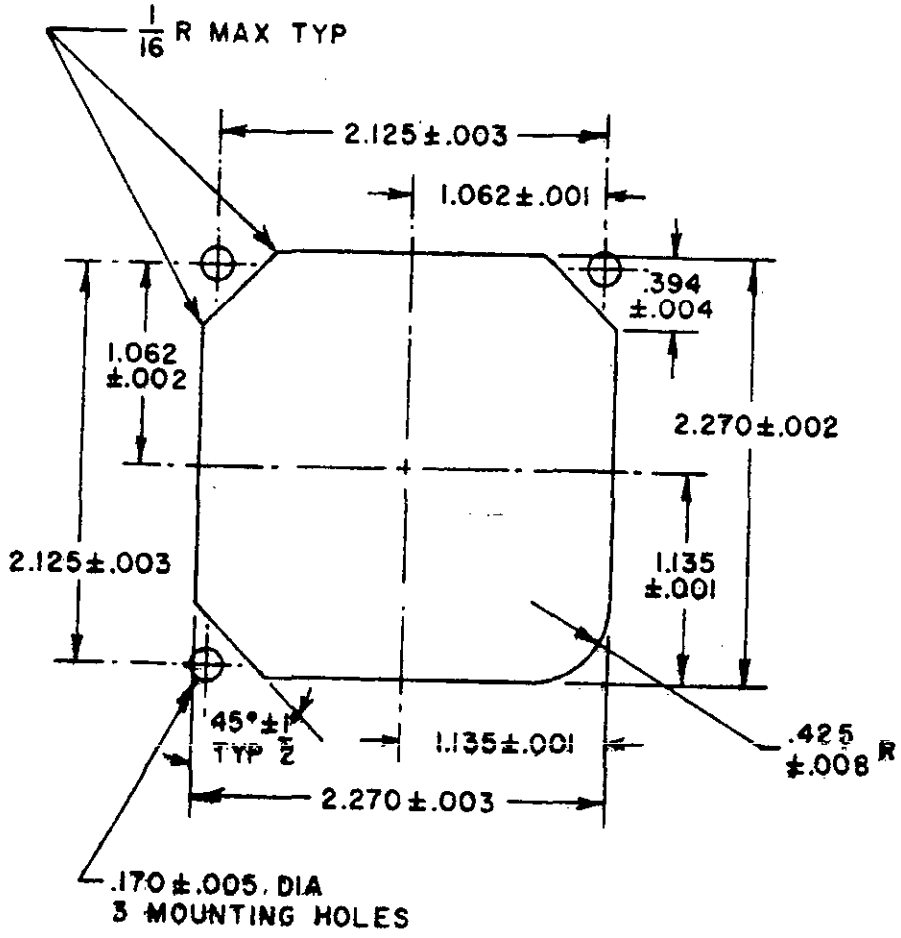
3.5.2.1 Longevity. The indicator shall have a minimum operating life (longevity as defined in MIL-STD-781) of 5,000 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF.

3.6 Performance. The indicator shall operate as specified herein after subjection to the following conditions:

a. Temperatures: Temperatures ranging from -54° to +71°C

b. Humidity: Relative humidity up to 95 percent

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:31Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/64

FIGURE 1. PANEL CUTOUT

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:31Z
Check the source to verify that this is the current version before use.

c. Altitude: Pressures ranging from 30 inches Hg (approximately sea level) down to 1.31 inches Hg (approximately 70,000 feet)

d. Salt fog: Salt sea atmosphere for a period of 50 hours

e. Vibration:

(1) Error - Roll and pitch oscillation and variation errors, when operating at any specified indication, not exceeding 1/8° when vibrated at an amplitude of 0.003 to 0.005 inch while the frequency is varied from 5 to 50 Hz

(2) Failure - Vibration at a double amplitude of 0.10 inch from 5 to 14 Hz and vibration of ±1g from 14 to 50 Hz

f. Fungus: Fungus growth as encountered in tropical climates

g. Sand and dust: Sand and dust particles as encountered in desert areas

h. Acceleration: Acceleration force of 5g with power applied, and an acceleration force of 20g with no power applied for a period of 1 minute

i. Temperature shock: Three cycles of temperature shock, each cycle consisting of subjecting the indicator to +71°C for 4 hours and then to -54°C for 4 hours

j. Rain: Rain as encountered in any locale.

3.6.1 Dielectric strength. The indicator shall withstand 500V d-c or 500V 60-Hz a-c power applied between isolated pins and between pins and the case or connecter shell for a period of 10 seconds.

3.6.2 Power variation. The indicator shall operate properly and continuously from input power ranging between 103 and 127V at any frequency from 320 to 480 Hz, or any combination thereof, and under the transient variations specified in MIL-STD-704.

3.6.3 Magnetic effect. The indicator shall not cause a compass deflection in excess of 5° under the conditions specified in 4.6.16.

* 3.6.4 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 for class ID equipment.

3.6.5 Sealing thermal shock. The indicator shall withstand 8 cycles of alternate immersions in tap water at 85° ±5°C and then at 5° ±4°C for 30 minutes at each temperature with not more than 5 minutes between each immersion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Bank angle dial. The bank angle dial shall be in accordance with figure 2 or 3, as applicable. The scale markings of the dial shall be color No. 37875 of FED-STD-595.

* 3.9 Case. The case shall conform to figure 4 and shall be hermetically sealed. The case shall be of such strength that the indicator will operate satisfactorily at an absolute pressure of 28 psi a temperature of 25°C. The case shall be so designed that the entire mechanism can be removed from the case and replaced without the use of special tools.

* 3.9.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium, or 95 percent helium and 5 percent carbon dioxide. Nitrogen shall be in accordance with BB-N-411. The absolute pressure of the filling medium in the case shall be 1 atmosphere.

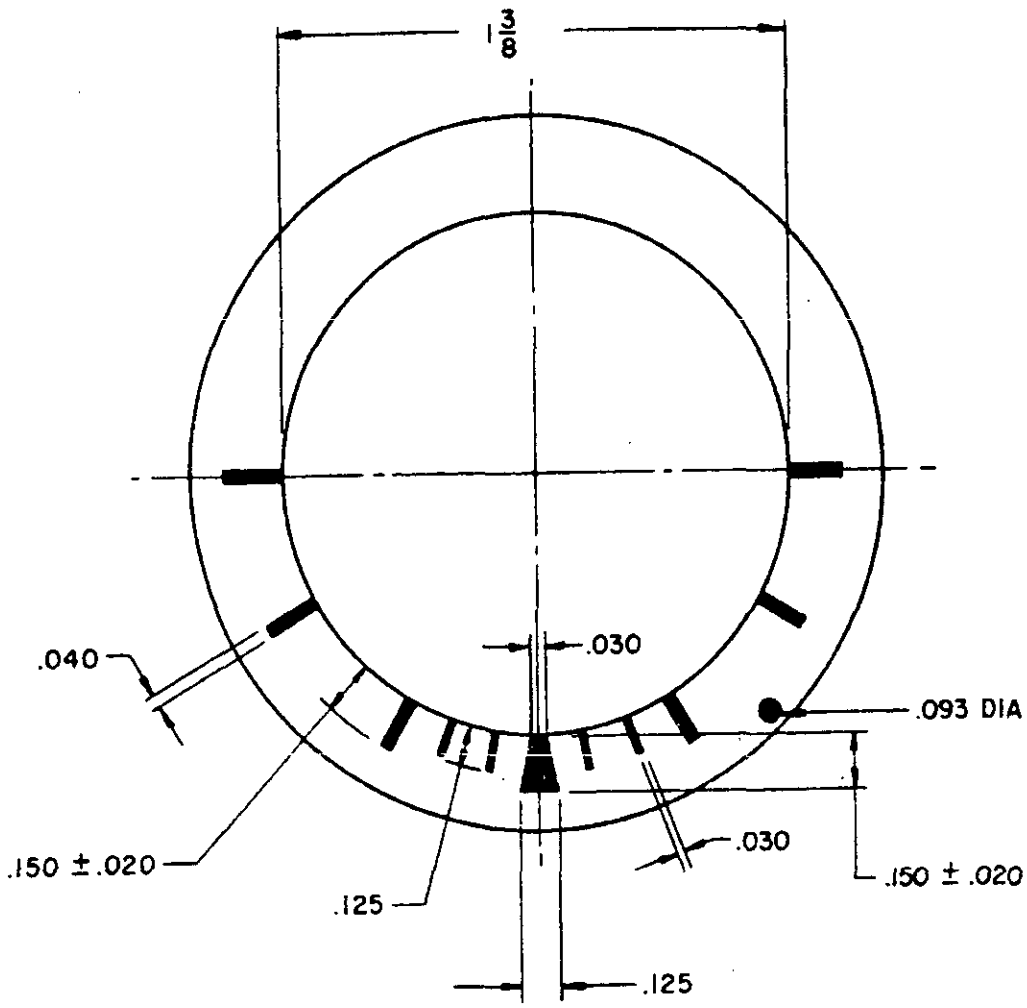
3.9.2 Leak rate. The indicators shall be so sealed that the maximum leak rate will not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential equivalent to the absolute pressure specified in 3.9.1.

3.10 Cover glass. The cover glass shall be flat and free from discolorations, scratches, and striae which interfere with normal reading. It shall be mounted in the indicator bezel by inserting it from the rear of the mounting bezel. The cover glass shall have a minimum thickness of 0.115 inch and a maximum thickness of 0.200 inch.

3.11 Lighting. The wedge technique shall be used to provide integral lighting. Supplemental lighting for the spheroid may be obtained by transillumination from within the spheroid. Integral lighting shall meet all requirements of MIL-L-25467 and MIL-I-27160; however, the gray portion of the spheroid shall at no time exceed 0.6 of the minimum brightness of the fixed white airplane.

* 3.11.1 Alternate lighting. The indicator shall be of such design that it may be lighted satisfactorily by an external post-type fixture.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.



MS27150-1 OR MS27150-3
(RED & WHITE LIGHTED)

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED,
TOLERANCES:

DECIMALS ± .005
FRACTIONS ± 1/64

FIGURE 2. Bank Angle Scale Details (Lower Location)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

3.11.2 Lighting pins. Power for lighting in accordance with MIL-L-25467 shall be derived from pins K and L. Power for lighting in accordance with MIL-L-27160 shall be derived from pins T and L. The lighting leads shall not be grounded to the indicator case.

3.11.3 Resistors. Although resistors shall not be permitted in the original design of the lighting system, space shall be allocated for incorporating a trimming resistor in each lighting circuit within the indicator. When resistors are used, they shall be in series with the entire lighting circuit.

3.11.4 Prime standard. Using parts that are identical to parts to be used in production indicators, a prime standard indicator shall be constructed that meets all display requirements of MIL-L-25467 and MIL-L-27160. Except for parts that affect the lighting design, the prime standard indicator may be a lighted model. The indicator and the recorded lighting data shall be submitted to the procuring activity for approval and if approved, it shall be certified as a prime standard.

3.11.4.1 Prime standard life. A minimum of 12 averge areas covering all representative areas of the face of the prime standard indicator shall be measured for brightness and color prior to submission and at least every 50 hours of operation thereafter. A change of ±20 percent brightness or ±5 percent color in two or more of these areas, using the initial data as a reference, will void the prime standard certification.

3.12 Antireflection coating. All reflecting glass surfaces of the cover glass and lighting wedge shall be covered with a reflection-reducing coating that meets the requirements of MIL-C-14806 in addition to withstanding the environmental conditions specified herein.

3.13 Sphere-bezel dimensions. The shortest distance from the sphere to the plane that contains the front edge of the indicator bezel shall be 0.350, +0.050 -0.200 inch.

3.14 Markings. All markings shall be clear and sharp. Dimensions shall be as specified and on any given mark, the width shall not vary noticeably throughout the length of the mark unless otherwise specified. All markings which, for some given indication, become adjacent and which are specified to have the same basic dimension at the adjacent point shall not be noticeably different at that point.

3.15 Miniature airplane. The miniature airplane shall be in accordance with figure 5. The rear edge of the miniature airplane shall be within 1/32 inch of the spheroid surface. The wings shall follow the curvature of ths spheroid. The top edge of the wings and the center dot shall be tilted back approximately 5° to facilitate night lighting. The miniature airplane shall be fixed directly to the wedge provided for integral lighting at the bottom two points of the "W",

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

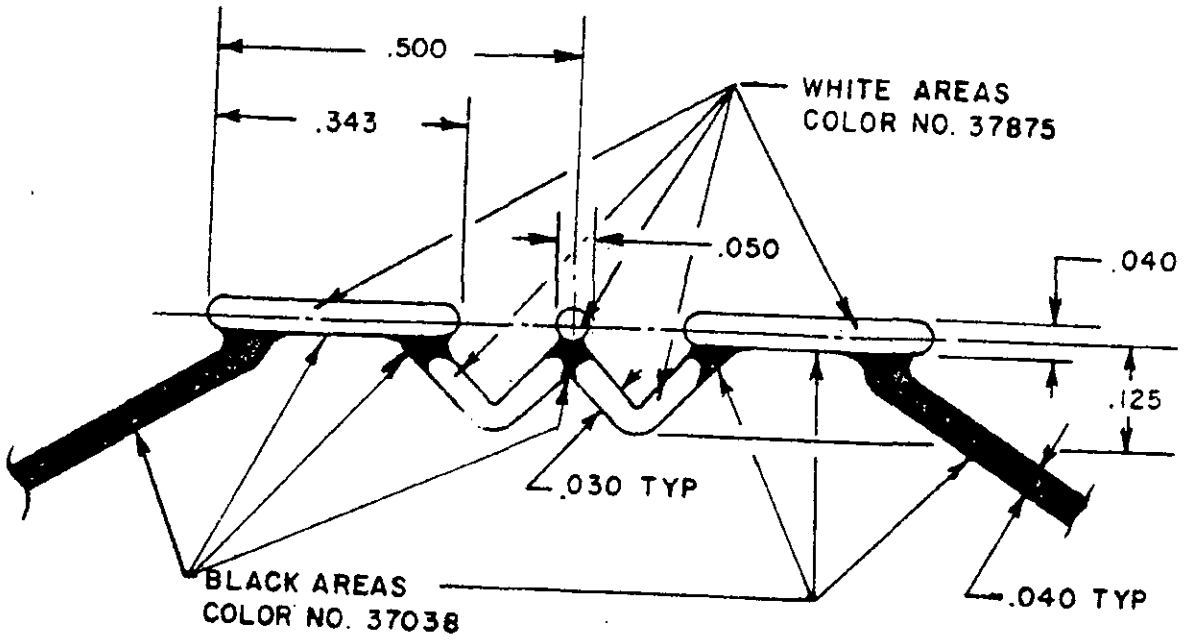
or attached with struts as shown on figure 5. The attaching areas shall not exceed 1/16 inch diameter each and shall be color No. 37038. The white areas of the miniature airplane shall be color No. 37875. The airplane shall have a black border in accordance with figure 5. Unless otherwise specified, all colors shall be in accordance with FED-STD-595.

3.16 Bank angle index. The bank angle index shall be in accordance with figure 5 and shall be affixed to rotate with the roll gimbal. The index shall not move with pitch deviations. The front surface of the bank angle index shall be in a plane that is parallel to the cover glass or shall be beveled as approved by the procuring activity. If parallel, this plane shall be in the same plane as the front surface of the small diameter of the bank angle dial and within ±3/64 inch. If beveled, the angles between the face of the index and the cover glass shall be approved by the procuring activity. At zero roll indication, the bank angle index shall be aligned with and radially within ±3/64 inch of the zero roll marking on the bank angle dial. The index shall be color No. 37875.

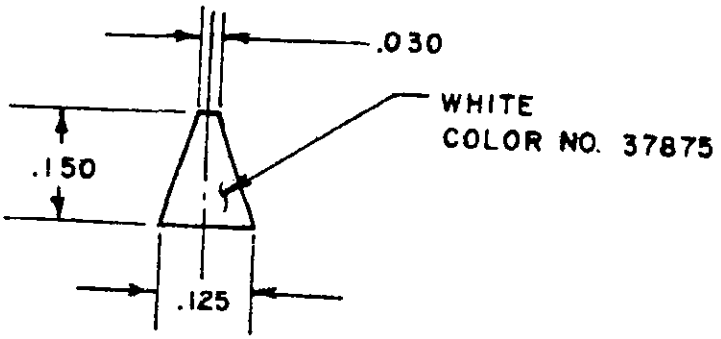
3.17 Spheroid. The spheroid shall be mounted in the roll gimbal and shall rotate with the gimbal about the roll axis for roll indications and rotate in relation to the roll gimbal about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator and the pitch axis shall be horizontal when the indicator is level and indicating a zero roll attitude. The horizon line on the spheroid shall be fixed to the spheroid at the zero pitch position. The spheroid shall be driven in pitch in such a way that for a dive maneuver, the horizon line will go above the miniature airplane and for a climb maneuver, the horizon line will go below the miniature airplane. The spheroid shall be in accordance with figure 6. The surface of the spheroid shall move 0.026 inch (nominal) for each degree of pitch change. The markings shall be in accordance with figure 6.

3.17.1 Markings and colors. Above the horizon line, the spheroid shall be light gray, color No. 36373, and all markings shall be black, color No. 37038. The horizon line shall consist of alternate black and white dashes, color No's 37038 and 37875, respectively, as shown on figure 6, with the center-most dash black and bisected by the vertical reference line. Below the horizon line, the

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:31Z
Check the source to verify that this is the current version before use.



MINIATURE AIRPLANE

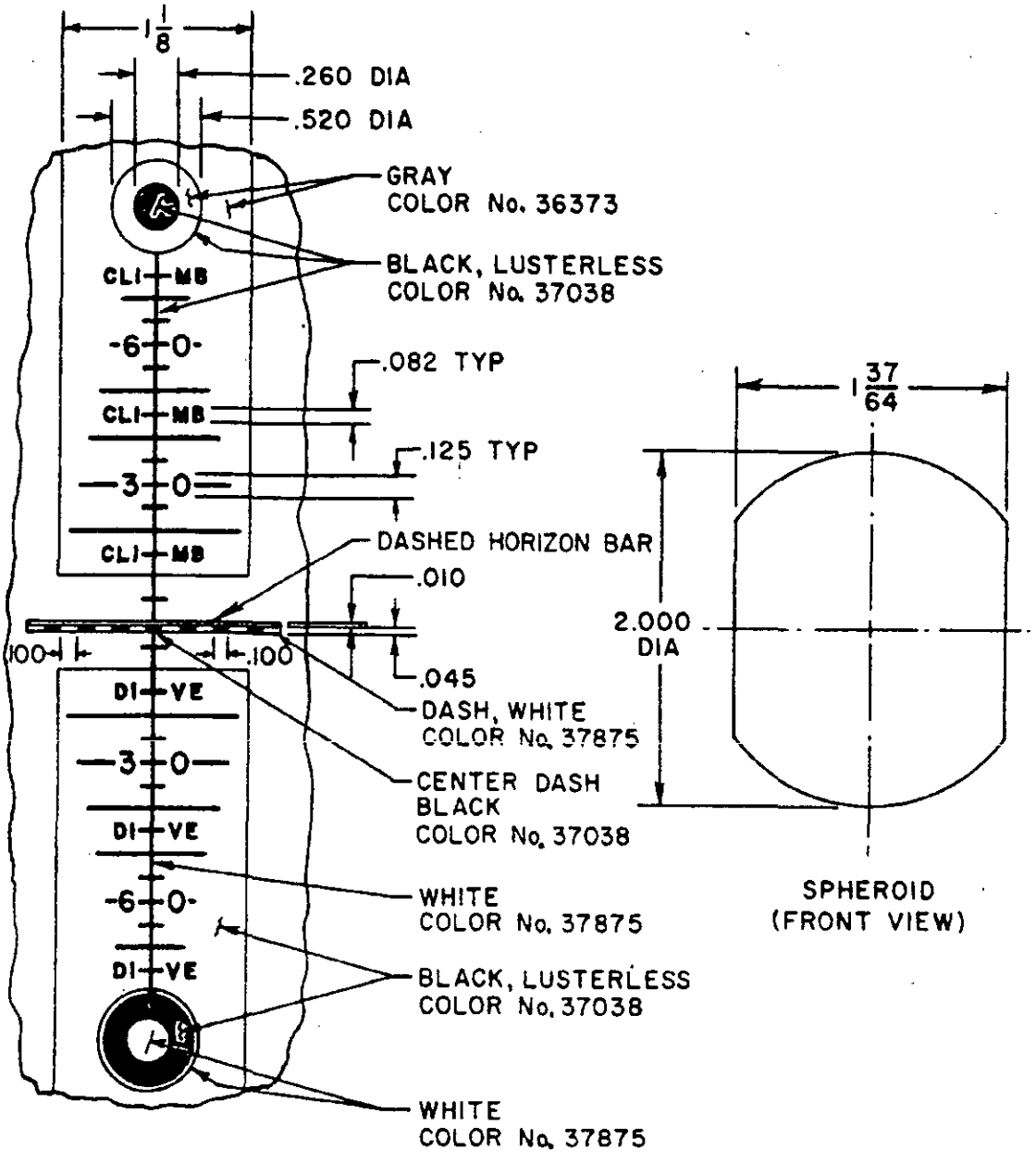


BANK ANGLE INDEX

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES : DECIMALS ±.005

FIGURE 5. Miniature Airplane and Bank Angle Index

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:31Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/64
DECIMALS ± .005

FIGURE 6. Spheroid Dimensions and Markings

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

spheroid shall be black, color No. 37038, and all markings shall be white, color No. 37875. The length of the major pitch lines shall be as follows:

0° (Horizon line)	1-1/2 inch
10°	1-1/8 inch
20°	1-1/32 inch
30°	29/32 inch
40°	25/32 inch
50°	21/32 inch
60°	17/32 inch
70°	13/32 inch
80°	Circle
85°	Solid dot.

The center of the solid dots shall be the ±90° pitch position and from center-to center shall represent 180° of pitch indication. The pitch indication shall be linear over the full range (180°) which shall be spaced over 270° of the spheroid surface.

3.18 Pitch trim. A sealed pitch-trim potentiometer shall be incorporated for adjusting the position of the spheroid. It shall be externally mounted on the hermetically sealed case on the lower right-hand side and shall be marked in accordance with figure 4. The reference mark on the knob shall extend across the end of the knob and shall continue along the side to a point on the knob which is adjacent to the index mark on the bank angle scale at the zero pitch-trim position. The body color of the knob shall be gray, color No. 36373, the reference arrow on the knob shall be black, color No. 37038, and the reference index mark on the bank angle scale shall be white, color No. 37875. The knob shall have approximately one-half turn of freedom in the clockwise direction and one-fourth turn of freedom in the counterclockwise direction from the zero pitch-trim position. The freedom in each direction shall be limited by a positive stop. When the knob is rotated to the stop in the clockwise direction, the spheroid shall rotate to deflect the horizon line upward to indicate a dive of between 10° and 20°. When the knob is rotated to the stop in the counter-clockwick direction, the spheroid shall rotate to deflect the horizon line downward to indicate a climb of between 5° and 10°. The servo system shall operate properly for any pitch trim adjustment. A minimum of 2 inch-ounces and a maximum of 6 inch-ounces of torque shall be required to rotate the pitch trim knob through the full range of the settable indications in both clockwise and counterclockwise directions. When two indicators are connected to the same transmitter, rotating the pitch trim knob of one through its range of movement shall not change the indication of the other indicator by more than 1/2°.

3.19 Attitude actuation. The spheroid shall be rotated about the pitch and roll gimbal axes by means of geared servo motors until a null is attained between the control transformer synchros on the indicator axes and the transmitter

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.

synchros in the pitch and roll displacement reference. The freedom of the spheroid rotation shall be unlimited about the roll axis and shall be a minimum of ±90° of indication about the pitch axis. Stops shall be incorporated in pitch so that the indication will not become unsynchronized with the input signal and so that the indicator will not become damaged when the input pitch signal exceeds the indicator ±90° pitch range and comes within the range again by continued rotation in the same direction.

3.19.1 Indicator synchro control transformer. The synchro control devices on the indicator axis shall be high impedance Eclipse-Pioneer type AY500-5 synchros, or equal. The servo circuit shall be so designed that when one to four indicators per axis are connected to a single test transmitter synchro, no noticeable change in performance can be noted. All synchro control device stator leads shall be electrically isolated from each other, from other circuitry in the case, and from the case.

3.20 Standard test transmitter. The standard test transmitter synchro shall be an Eclipse-Pioneer type AY201-1, or equal, with minimum and known errors. The errors shall be subtracted from those of the indicator under test.

3.20.1 The transmitter shall be equipped with an accurate dial having a ±1° vernier and readable to 0.1°. It shall be graduated through 360° and shall be capable of continuous rotation.

3.20.2 Calibration. The test transmitter shall be calibrated and the rotor and stator leads identified as follows:

a. Designate rotor leads as H and C, with C grounded

b. Designate one stator lead as Z and connect it to C

c. Apply 26V 400 Hz across H and C and measure the voltage across the two remaining stator leads and the voltage across H and one of the stator leads

d. Turn the rotor until the voltage across H and the one stator lead is maximum and the voltage across the two stator leads is minimum. Lock the dial to the rotor with the dial exactly on zero

e. Turn the dial through increasing numbers and measure the voltage across the two undesignated stator leads and Z. The voltage across XZ will increase before decreasing and across YZ will decrease before increasing.

3.20.3 Test transmitter excitation. The test transmitter rotors shall be excited from the same power supply voltage which operates the indicator. It shall be so applied that the voltage from rotor lead C to rotor lead H will be in phase with the voltage from pin A to pin B of the indicator with lead C and pin A common as shown in table I.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the currentversion before use.

3.20.4 Indication zeroing and sensing. When the pitch and roll servo control transformers of the indicator are connected to standard test transmitters in accordance with table I, with the respective transmitters set at the 180° dial reading, the indications shall come to rest at zero. Increasing dial readings of the respective test transmitter shall produce right-roll and pitch-up indications. Decreasing dial readings shall produce opposite indications.

3.21 Followup operation. The electrical and mechanical design of the indicator shall be such that in the process of followup, the indications in pitch and roll shall be smooth without hunting, oscillating, or jumping greater than 1/8° total amplitude in pitch and 1/8° total amplitude in roll when connected to a standard transmitter. There shall be no sustained electrical or mechanical oscillations in either axis greater than 1/8° total amplitude for any indication. The servo mechanisms shall be 70 to 100 percent damped and shall not lag the signal source by more than 10° at maximum rate of servo operation.

3.21.1 Followup rate. The rate at which the motors drive the gimbals shall be not less than 300° per second in roll nor less than 180° per second in pitch.

3.21.2 Followup accuracy. The pitch and roll indications shall follow the input signals from their respective test transmitter synchro to within 1° for positions 0° to ±30° in roll and pitch and to within ±2° for all other positions.

3.22 Amplifier channels. The indicator shall incorporate rugged, reliable two- or three-stage transistor or magnetic amplifiers which shall have push-pull output circuitry. The indicator shall continue to operate, with reduced performance, should partial failure of the amplifier occur. Failure of one amplifier channel shall have no effect on the operation of the other channel.

3.23 Zeroing adjustments. Means shall be incorporated to externally adjust the position of the spheroid ±3° minimum in pitch and roll. The initial setting of the pitch and roll indication at the time of delivery of the indicator shall indicate zero pitch and roll within ±1/2° when the indicator is connected to a test transmitter set at electrical zero. The pitch adjustment shall be combined with the pitch trim circuit by means of a settable knob. The roll adjustment shall be on the rear of the case and shall be adequately marked. These adjustments shall be locked in place once they are set.

3.24 Power failure indication. A power failure flag indicator conforming to MS27150 shall be incorporated into the lower left of the face of the indicator. A lead shall be brought out through the electrical connector through pin J, and the circuit shall be such that when this pin is connected to pin A and power is applied, the warning will disappear. When the connection is broken or power is removed, the warning shall appear. The design shall be such that the warning will also appear due to electrical failure within the indicator insofar as practical. The word OFF shall be black, color No. 37038, on a red background and shall be illuminated under night lighting conditions in accordance with 3.11.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.

TABLE I. Indicator Connections

Electrical Pins	Connections
A	Grounded to neutral of indicator (not grounded to case)
B	115V, 400-Hz single-phase power
C	No connection
D (X) winding E (Y) winding F (Z) winding	Roll servo control transformer stator leads
R X winding H Y winding P Z winding	Pitch servo control transformer stator leads
J	Power failure warning (connect to pin A to complete circuit)
G	Grounded to case (MS27150-5 only)
K L	Red internal lighting power input (common to lighting circuit)
M	Pitch null test terminal (pitch error voltage between this pin and pin A)
N	Roll null test terminal (roll error voltage between this pin and pin A)
T L	White internal lighting common (not connected to case)
V, U, S	No connections

3.25 Electrical connector. The electrical connector on the indicator shall be a Bendix Scintilla type as shown on figure 4, or equal, and shall be affixed to the case as shown thereon (widest keyway at 90° position). The pins on the connector shall have a minimum of 0.00005-inch gold plate over either a 0.0003-inch silver plate in accordance with QQ-S-365 or a 0.00015-inch nickel plate. The connector shall mate with Bendix clamp type PT06A-14-19S or potting type PT06P-14-19S receptacle, or equivalent. The electrical connections to the

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:31Z
Check the source to verify that this is the current version before use.

connector on the indicator shall be in accordance with table I. The electrical connector shall be equipped with a cap or plug and gasket in a manner that will protect the electrical surface from water vapor.

3.26 Electrical power. The power supply to the indicator shall be 115V 400-Hz single-phase in accordance with MIL-STD-704. Power-input pin connections shall be in accordance with table I.

3.26.1 Power consumption. The power consumption shall not exceed 20 va.

3.27 Compatibility. The indicator shall operate as specified herein when connected in parallel with the circuit shown on figure 7.

3.28 Weight. The weight of the indicator shall not exceed 2.75 pounds.

3.29 Electronic parts. Electronic parts shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists shall not be used unless approved by the procuring activity.

3.29.1 Other components. All other components in the electronic system shall conform to applicable specifications where existent.

3.30 Numerals and letters. Unless otherwise specified, all numerals and letters on the indicator, other than the nameplate, shall be in accordance with MS33558.

* 3.31 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454.

3.32 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

* 3.33 Lubrication. The system shall be lubricated in accordance with MIL-STD-838.

3.34 Finishes and protective coatings

3.34.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.34.1.1 Small holes and case inserts need not be anodized.

3.34.1.2 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.34.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:31Z
Check the source to verify that this is the current version before use.

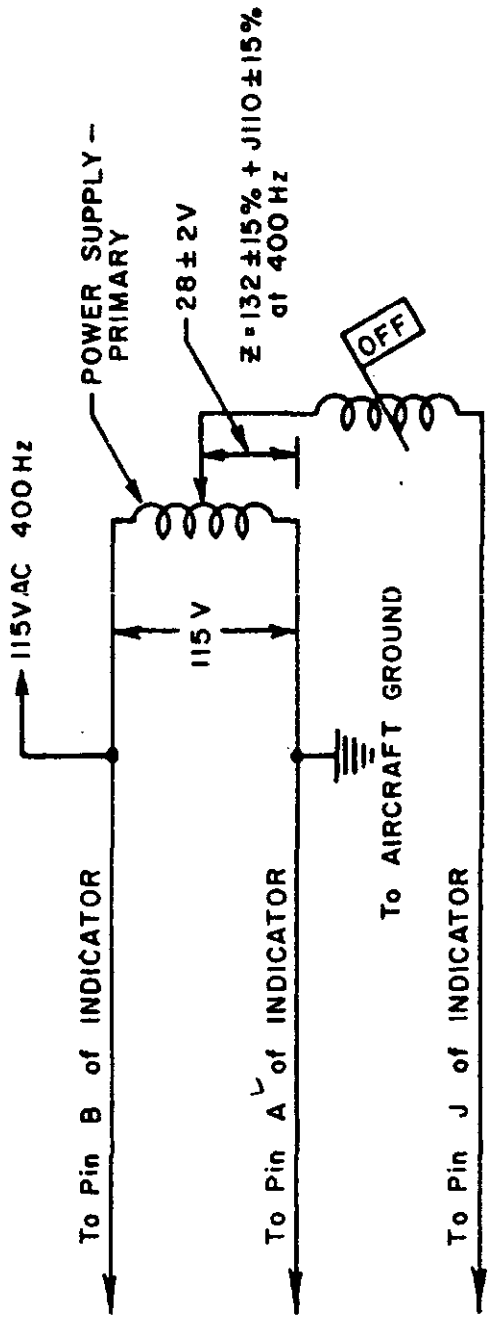


FIGURE 7. Power Warning Flag Compatibility Circuit

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:31Z
Check the source to verify that this is the current version before use.

3.34.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.34.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.34.1.6 Parts enclosed in a hermetically sealed container need not be anodized or plated.

3.34.1.7 When the part is plated with tin over a copper flash, the part need not be anodized.

3.34.1.8 When necessary for electrical bonding, parts need not be anodized.

3.34.2 Steel parts. Steel parts shall be cadmium plated, where practicable, in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.35 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.36 Workmanship. The indicator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, radii of fillets, marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring, and riveting, alignment of parts, and tightness of assembly screws and bolts.

3.36.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolernnces may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.36.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, or threads.

3.36.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.36.4 Gears. Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots, or other irregularities. Where required for accuracy adjustment, gear assemblies shall be free from backlash.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:31Z
Check the source to verify that this is the current version before use.

3.36.5 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, or other foreign material after assembly. Burrs and sharp edges as well as resin flash which might crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Qualification tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Unless otherwise specified, all tests shall be conducted at atmospheric pressure (apprximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Voltage input. Unless otherwise specified, the indicator shall be tested with 115V 400-Hz (nominal) single-phase power.

4.3.3 Proper connection. When it is specified that the indicator will be properly connected, it shall be connected to a test fixture which provides a lighting circuit voltage variable from 5V (ac or dc) to approximately 0V, synchro signals in accordance with 3.20 and 3.21, 0 a-c power as required, and pin J shall be jumped to pin A through a closed circuit.

4.3.4 Lighting. All indicators shall be visually checked against the prime standard specified in 3.11.4. Production indicators shall also be checked in accordance with all the tests specified in MIL-L-25467 and MIL-L-27160.

4.3.4.1 Visual observation. Unless otherwise specified, all visual observations of indicator display shall be made at a distance of approximately 2 feet at any point within a sighting cone 30° from the normal to the surface of the cover glass.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

4.3.5 Standard test transmitter. The standard test transmitter used to test the indicator shall be an Eclipse-Pioneer type AY-201-1, or electrical equal, precision synchro with minimum and known errors. The rotor shall be settable and shall rotate through 360°. The settings shall be readable to within 6 minutes of arc and each scale marking in degrees shall be equal to 1° of synchro rotor rotation.

4.4 Qualification testing

* 4.4.1 Test samples. The test samples shall consist of a minimum of six indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. The samples submitted to the procuring activity shall pass the sampling plan A tests during spot checks by that activity.

4.4.2 Test report. When tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity:

a. Three copies of a test report in accordance with MIL-STD-831

* b. The three samples that were tested.

4.4.3 Qualfication. The qualification tests shall consist of all the tests described under 4.6 except 4.6.43 and 4.6.44.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Early failure detection

c. Starting

d. Pitch and bank zero

e. Pitch trim

f. Zeroing adjustments

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:31Z
Check the source to verify that this is the currenversion before use.

g. Sensitivity

h. Followup rate

i. Followup accuracy

j. Gimbal freedom

k. Followup operation

(1) Indication lag

(2) Hunting and jumping

(3) Overshooting

l. Leak rate

m. Power warning indicator

n. Lighting (visual comparison)

o. Fogging.

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. Three indicators shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6.

a. Individual tests (except 4.6.2)

b. Magnetic effect

c. Acceleration (power on)

d. Dielectric strength

e. Vibration error

f. Horizontal line and bank angle index null position shift

g. Low temperature operation

h. High temperature operation

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:31Z
Check the source to verify that this is the current version before use.

i. Power variation

j. Indication lag

k. Power consumption

l. Weight

m. Component compatibility.

* 4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), 3 indicators shall be selected at random from the first 15 on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests (except 4.6.2)

b. Electromagnetic interference

c. High temperature exposure

d. Temperature altitude

e. Temperature shock

f. Pressurization

g. Sealing thermal shock

h. Humidity

i. Fungus

j. Rain

k. Salt fog

l. Dust

m. Shock

n. Vibration failure

o. Acceleration (power off)

p. Lighting (measured)

q. Production reliability.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.

Note: After completion of the sampling plan A tests, at the option of the procuring activity, the tests may be divided into three groups as follows, to be conducted on three sets of three indicators if a need exists to expedite testing. Each indicator shall meet all individual tests after completing the group of specific tests to which subjected.

Group I	Group II	Group III
Electromagnetic interference	Sealing thermal shock	Shock
High temperature exposure	Humidity	Vibration failure
Temperature-altitude	Fungus	Acceleration (power off)
Temperature shock	Rain	Lighting (measured before and after reliability)
Pressurization	Salt fog	Production reliability.
	Dust	

4.5.2.3 Rejection and retest. When one item from a production run fails to meet the specification, no items on hand or later produced shall be accepted until the extent and cause of failure are determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Early failure detection. The indicator shall be properly connected and operated for a total of 50 hours so that any early failure of the indicator may be detected. The total operating time shall be achieved in 25 cycles, each cycle consisting of a 2-hour period of operation followed by a 5-minute period during which no power shall be applied. During the operating periods, both axes of the attitude display shall be oscillated simultaneously through ±30° at a rate of 5

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.

to 7 times per minute, the lighting circuit shall be energized with 5V ac or dc, and the power warning flag system shall show that the indicator is in operation. During the resting or cooling-off periods, all power shall be removed from the instrument under test, including the power to the lighting and flag circuits, and the standard test transmitter shall remain excited and its rotor shall continue to be displaced through ±30° at a rate of 5 to 7 times per minute. There shall be no degradation in indicator performance. This test shall be conducted only once on each indicator.

4.6.3 Starting. The indicator shall be properly connected and power shall be applied. Upon application of power, the warning flag shall disappear immediately and the indicator shall be ready for operation.

4.6.4 Pitch and bank zero. The standard test transmitters in the pitch and roll axes shall be set at the electrical zero position specified in 3.18. Power shall be applied and the indicator shall follow up to the zero indications in pitch and bank within ±1/2° from any random indication in either axis. When the indication is stable under these conditions, the error signal voltage from the two control transformers shall be as near as practical to minimum voltage. This test shall be repeated and the indication shall return to within ±1/4° of its previous indication.

4.6.5 Pitch trim. The pitch trim knob shall be adjusted to its limits of one-fourth turn counterclockwise and one-half turn clockwise and the deflection of the horizon observed. The respective deflections shall be from 5° to 10° down and from 10° to 20° up from the level flight indication. With the pitch trim knob in the zero pitch trim position, the indicator shall indicate zero pitch within plus or minus one-half of the horizon line width. The torque required to rotate the pitch trim knob shall be measured throughout the range of the settable indications in both the counterclockwise and clockwise directions and shall be within 2 to 6 inch-ounces. Two standby indicators of the type specified herein shall be connected to the same transmitter and when the pitch trim knob is moved through its range of motion, there shall be not more than 1/2° change in the indication of the other indicator.

4.6.6 Zeroing adjustments. The indicator shall be properly connected to the standard test transmitters each of the which shall be set to electrical zero. The roll zeroing adjustment shall be moved to its extremes and the deflection of the bank angle index noted. This deflection shall be not less than ±3° from the zero position. The pitch test transmitter shall be rotated 3° in climb and then 3° in dive. In each case, the pitch trim knob shall be loosened on its shaft and the shaft rotated until the indication is zero. The pitch trim knob shall then be tightened to the shaft at the zero position and shall meet the test specified in 4.6.5. This test shall be performed during the initial individual testing and during the final testing conducted as a part of the sampling plan A testing following the reliability test.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:31Z
Check the source to verify that this is the current version before use.

4.6.7 Sensitivity. The test transmitter for the roll axis shall be moved 1/4° in right bank and returned to the original position and 1/4° in left roll and returned to the original position. With the pitch trim knob in the zero pitch trim position, the test transmitter for the pitch axis shall be moved 1/4° in dive and returned to the original position and 1/4° in climb and returned to the original position. There shall be a smooth perceptible movement of the indicator bank angle index and horizon line for each movement of the appropriate test transmitter. This test shall be repeated in dive and climb with the pitch trim knob in each of its extreme positions, except the test transmitter shall be moved 1/2° in each direction.

4.6.8 Followup rate. The indicator shall be properly connected to a romote signal source or standard test transmitter. The followup rates in pitch and roll shall be determined. The computed rate in the roll axis shall be not less than 300° per second and the computed rate in the pitch axis shall be not less than 180° per second.

4.6.9 Followup accuracy. The rotor of the standard test transmitter in the roll axis shall be displaced to 5 positions other than the zero position. The indication of the indicator shall position itself with the rotor of the standard test transmitter within ±1° for each position between 0° and 30° left and right bank and within 2.0° for any position over the remaining 300°. The rotor of the standard test transmitter in the pitch axis shall be displaced to 5 positions other than the zero position. The indication of the indicator shall position itself with the rotor of the standard test transmitter within ±1° for each position between 0° and 30° dive or climb and within 2.0° for any position between 30° and 90°.

4.6.10 Gimbal freedom. The indicator shall be properly connected to the romote signal or standard test transmitter and the transmitter positioned ±360° in roll and ±90° in pitch. The indicator shall follow smoothly without noticeable sticking, overshooting, or hunting. The pitch axis test transmitter signal shall be rotated through 360°. The pitch indication shall synchronize with the input signal when it is within ±90° range of the pitch indication and the indicator shall not be damaged by the input signal exceeding the ±90° pitch indication range.

4.6.11 Followup operation. Suitable test equipment shall be connected to the appropriate test pins of the indicator to observe and record any hunting, jumping, lag, or overshooting of the indication when the indicator is made to follow the standard transmitter at various rates in both the pitch and roll axes. This shall be possible with the indicator sealed. The recorded data shall be accomplished as follows: The data shall be recorded during the initial individual tests preceding all other tests and during the final individual tests following the completion of all other tests. While performing the individual tests at other times, the indicator may be checked visually for hunting, jumping, lag, or overshooting.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.

4.6.11.1 Indication lag. The indication in the pitch axis shall not lag behind the pitch remote input transmitter by more than 5° for a shaft speed of 90° per second nor more than 10° for a shaft speed of 180° per second. The indication in the roll axis shall not lag behind the roll remote input transmitter by more than 5° for a shaft speed of 90° per second nor more than 10° for a shaft speed of 300° per second.

4.6.11.2 Hunting and jumping. Hunting and jumping of the indicator display in pitch and roll, with no inputs from the test transmitters, shall not have an amplitude greater than 1/8°. The manufacturer shall conduct a survey on the indicator of his design operated over the full range of operating speeds. The testing rates chosen for hunting and jumping shall be those that produce the most erratic motion and shall include at least one rate within each of the following ranges:

Pitch transmitter - 0° to 20°, 20° to 60°, 60° to 90°, 90° to 120°, and 120° to 180° per second

Roll transmitter - 0° to 20°, 20° to 60°, 60° to 90, 90° to 120°, 120° to 180°, 180° to 240°, and 240° to 300° per second

The peak-to-peak amplitude of the hunting and jumping shall not exceed 1/8° for shaft speeds between 0° and 20° per second in pitch and roll, 1/4° for shaft speeds between 20° and 90° per second in pitch and roll, and 1/2° for shaft speeds for the remaining rates in both pitch and roll.

4.6.11.3 Overshooting. Upon completion of pitch and roll maneuvers, overshooting shall be held to a practical minimum and shall not exceed one overswing with a maximum amplitude of 3° with a step equal to the maximum allowable followup lag.

4.6.12 Leak rate. The indicator case shall be tested for leaks by means of a mass-spectrometer type helium leak detector or other suitable means approved by the procuring activity. The leak rate shall not exceed that specified in 3.9.2.

4.6.13 Power warning indicator. The indicator shall be properly connected with no power applied. Power shall be applied and the warning shall disappear. The power shall be disconnected and the warning shall immediately appear. Power shall be reapplied and when jumper pin A to J is disconnected, the warning shall appear.

4.6.14 Lighting (visual comparison). Both red and white lighting circuits shall be tested by visual comparison of the production indicator against the prime standard. The tests shall be conducted as follows:

4.6.14.1 Color. For indicators using red lighting, 5V shall be applied to the indicator being tested and to the prime standard and for indicators using white

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

lighting, 4.5V shall be applied. In each case, the indicator under test shall be visually compared with the prime standard and the two shall appear identical in color.

4.6.14.2 Brightness. The prime standard and the production indicator under test shall be placed side by side at eye level. The two indicators shall be viewed at a distance of approximately 2 feet from any point within a sighting cone 30° from the normal to the surface of the cover glass. The overall lighting of the two units shall be compared with 3.00 ±0.030V applied to the red lighting terminals of the indicator under test. General overall lighting level comparison shall then be made between the indicator being tested and the prime standard with 2.8V applied and then varied up to 3.2V. If the general overall lighting level of the indicator being tested does not fall within the brightness limits of the prime standard when energized at the specified voltage levels, the indicator shall be rejected. This test shall be repeated for indicators utilizing the white lighting.

4.6.14.2.1 Indicators that are rejected due to the test specified in 4.6.14.2 shall be subjected to all display tests specified in MIL-L-25467 and MIL-L-27160, as applicable. For indicators that are tested in accordance with MIL-L-25467, the brightness of the entire white area of the miniature airplane shall be within 0.6 and 1.8 foot-lamberts.

4.6.15 Fogging. The indicator shall be placed in a 71° ±2°C controlled ambient temperature and energized for a minimum period of 1 hour. At the end of the 1-hour period and while still at the specified temperature, an ice cube shall be rubbed on the face of the indicator glass for a period of 1 minute ±5 seconds. The glass shall be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of water or oil fog. Evidence of fogging on the inner surface of the cover glass shall be cause for rejection.

4.6.16 Magnetic effect. This test shall be conducted first with no power applied to the indicator and then with the indicator operating on rated power. The indicator shall be held in various positions magnetically east or west of and 5 inches from the center of a free magnet approximately 1-1/2 inches long in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted. The maximum deflection of the magnet shall not exceed 5°. An aircraft compass with the compensating magnet removed may be used as the free magnet.

4.6.17 Acceleration (power on). The indicator shall be properly connected with power applied and accelerated along its normal vertical axis with a force of 5 g's. The attitude display shall not vary by more than 1° in pitch or roll from the initial indication. This test shall be conducted with the attitude set at zero pitch and roll and with each axis displaced 90°.

4.6.18 Dielectric strength. A potential of 500V dc or 500V rms at 60 Hz shall be applied between isolated pins and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation. Insulation breakdown shall be considered as a leakage current in excess of 2.5 milliamperes.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.

* 4.6.19 Vibration error. The indicator shall be connected to the test transmitter and subjected to a vibration error test in accordance with MIL-STD-810, method 514, equipment category (b), table 514.1-II, procedure I, part 2, curve B, except the amplitude shall be from 0.005 to 0.003 inch at frequencies from 5 to 50 Hz. The indicator shall meet the following tests which shall each be conducted at three indications:

4.6.19.1 Roll and pitch oscillation. The maximum total amplitude of roll and pitch indicated oscillation throughout the frequency range shall be noted and shall not exceed 1/8°.

4.6.19.2 Roll and pitch variation. The maximum variation of roll and pitch indicated during vibration from the original indication before vibration throughout the frequency range shall be noted and shall not exceed 1/8°.

4.6.20 Horizon line and bank angle index null position shift. The indicator shall be properly connected to standard test transmitters in the pitch and roll axes. The transmitters shall be set to drive the indications in both axes to the zero position. The input voltage and frequency shall be varied within the following limits and combinations thereof:

Frequency	320 to 480 Hz
A-C voltage	103 to 127V

The horizon line shall not shift more than ±1/16 inch from normal null position. The bank angle index shall not shift more than one-half the narrow end of the pointer width from normal null position.

4.6.21 Low temperature operation. The indicator shall be properly connected to the standard test transmitter, but no power shall be applied, and placed in a chamber at a temperature of -54° ±2°C for a period of 4 hours. At the end of this period and while the indicator is at the reduced temperature, power shall be applied for a period of 10 minutes. At the end of the 10-minute period and while still at the reduced temperature, the indicator shall meet the test specified in 4.6.7 except as follows: The test transmitter for the roll axis shall be moved 1/2° in right bank and returned to the original position, and 1/2° in left bank and returned to the original position. With the pitch trim knob in the zero position, the test transmitter for the pitch axis shall be moved 1/2° in dive and returned to the original position, and 1/2° in climb and returned to the original position. There shall be a smooth percepticle movement of the indicator bank angle index and horizon line for each movement of the appropriate test transmitter. The test shall be repeated in dive and climb with the pitch trim knob in each of its extreme positions, except the test transmitter shall be moved 1° in each direction.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:31Z
Check the source to verify that this is the current version before use.

4.6.22 High temperature operation. The indicator shall be properly connected, placed in a chamber at a temperature of 71° ±2°C, and with power applied shall be operated for a period of 4 hours. At the end of this period and while at the high temperature, the indicator shall meet the tests specified in 4.6.4 and 4.6.7 except as follows:

a. 4.6.4: The indicator shall follow up to the zero indications in pitch and bank within ±1° from any random indication in either axis

b. 4.6.7: The test transmitter for the roll axis shall be moved 1/2° in right bank and returned to the original position, and 1/2° in left bank and returned to the original position. With the pitch trim knob in the zero position, the test transmitter for the pitch axis shall be moved 1/2° in dive and returned to the original position and 1/2° in climb and returned to the original position. There shall be a smooth perceptible movement of the indicator bank angle index and horizon line for each movement of the appropriate test transmitter. The test shall be repeated in dive and climb with the pitch trim knob in each of its extreme positions, except the test transmitter shall be moved 1° in each direction.

4.6.23 Power variation. The indicator shall be properly connected with power applied and subjected to the transient power conditions specified in MIL-STD-704 for a total of 10 cycles at a rate of 1 cycle per second. No damage shall result. The indicator shall then be subjected to the tests specified in 4.6.4, 4.6.7, 4.6.10, and 4.6.13 at voltages and frequencies of 103V 320 Hz, 103V 480 Hz, 127V 320 Hz, and 127V 480 Hz. The tolerance for the test specified in 4.6.4 shall be 3/4° for zero bank and for the test specified in 4.6.7 shall be 1° in roll and 3/4° in pitch.

4.6.24 Power consumption. The power consumption shall be measured with suitable meters and shall not exceed 20 va. ac. The power factor shall be maintained between 0.70 and 1.00.

4.6.25 Weight. The weight of the indicator shall be checked and shall not exceed 2.75 pounds.

4.6.26 Component compatibility. The indicator shall be subjected to a component compatibility test to determine compliance with 3.27. The indicator flag shall remain in view when connected as shown on figure 7, and when pins A and J are connected together, the flag shall disappear from view.

* 4.6.27 Electromagnetic interference. The indicator shall be subjected to electromagnetic interference tests in accordance with MIL-STD-462. The test methods shall be those specified for class ID equipment.

4.6.28 High temperature exposure. The indicator shall be properly connected and subjected to a high temperature exposure test in accordance with MIL-STD-810,

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.

method 501, procedure II, except the exposure period shall be 24 hours and no power shall be applied. At the end of the 24-hour period and while still at the high temperature, the indicator shall meet the test specified in 4.6.10.

4.6.29 Temperature-altitude. The indicator shall be subjected to temperature-altitude test in accordance with MIL-STD-810, method 504, for class 2 equipment, except that examination shall be performed at the end of 72 hours. At the end of this period, the pressure in the chamber shall be reduced to the equivalent of 70,000 ±500 feet and power applied. The time required for the indicator to start operating shall not exceed 5 minutes. There shall be no binding or sticking at any indication. After the indicator has returned to room temperature, it shall meet the individual tests.

4.6.30 Temperature shock. The indicator shall be subjected to a temperature shock test in accordance with method 503 of MIL-STD-810. At the conclusion of this test, the indicator shall meet the individual tests.

4.6.31 Pressurization. The indicator shall be placed in a chamber at an absolute pressure of 28 psi and a temperature of 25°C for a period of 4 hours during which the indicator shall not become deformed nor damaged. At the end of the 4-hour period, any evidence of deformation, damage, or unsatisfactory operation shall be cause for rejection. The indicator shall then pass the test specified in 4.6.12.

.32 Sealing thermal shock. The indicator shall be immersed alternately in tap water maintained at 85° ±5°C and 5° ±4°C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes. Not more than 5 seconds shall elapse between bath immersions. No damage to the hermetic seal shall result from this test. Following the immersions, the indicator shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The initial maximum detected leak rate at a differential pressure of 1 atmosphere shall not permit more than a 10-percent loss of the total filling medium after 1,000 hours.

4.6.33 Humidity. The humidity test shall be conducted in accordance with method 507 of MIL-STD-810, procedure I. No sticking or binding in the pitch trim knob shall result and there shall be no evidence of corrosion or rusting of the internal of external parts of the indicator, including the pitch trim knob mechanism.

4.6.34 Fungus. The indicator shall be subjected to a fungus test in accordance with method 508 of MIL-STD-810. No sticking or binding in the pitch trim knob shall result and there shall be no deterioration that would adversely affect subsequent operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

4.6.35 Rain. The indicator shall be subjected to a rain test in accordance with MIL-STD-810, method 506, except that the indicator shall be mounted for 1/2 hour with the dial in a vertical plane and 1/2 hour with the dial in a horizontal plane (dial face up) in the rain test chamber. Upon removal from the chamber, the indicator shall be examined and there shall be no sticking or binding in the pitch trim knob as a result of this test.

4.6.36. Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, for a period of 50 hours. At the end of the 50-hour period, there shall be no sitcking or binding in the pitch trim knob or roll trim potentiometer nor any damage to the indicator. The tolerances specified for the individual tests shall apply.

4.6.37 Dust. The indicator shall be subjected to a dust test in accordance with MIL-STD-810, method 510. Upon completion of this test, the indicator shall be removed from the chamber and allowed to cool at room temperature. There shall be no sticking or binding in the pitch trim knob as a result of this test.

* 4.6.38 Shock. The indicator shall be subjected to a shock test in accordance with MIL-STD-810, method 516, procedure I, figure 516.1-1 for flight vehicle equipment.

4.6.39 Vibration failure. With no power applied, the indicator shall be subjected to a vibration failure test in accordance with MIL-STD-810, method 514, equipment category (b), table 514.1-II, procedure I, part 2, curve B except that at frequencies from 5 to 14 Hz, the amplitude shall be 0.10 inch and at frequencies from 14 to 50 Hz, the acceleration shall be held at 1g. At the conclusion of this test, the indicator shall be subjected to and shall meet the tests specified in 4.6.4 and 4.6.7. There shall be no damage to any part of the indicator as a result of this test.

* 4.6.40 Acceleration (power off). The indicator, with no power applied, shall be mounted in its normal operating position and accelerated at 20g's for 1 minute in each axis, first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. At the end of the 1-minute acceleration period, the indicator shall meet the individual tests except 4.6.2. No damage to the indicator shall result from this test.

4.6.41 Lighting (measured). Lighting of the indicators shall be tested in accorddnce with MIL-L-25467 and MIL-L-27160, as applicable, both before and after the reliability test. The red and white lighting circuits shall each be operated for 1,500 hours either concurrently or consecutively.

* 4.6.42 Qualification reliability. The six indicators specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781, test level F,

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:31Z
Check the source to verify that this is the currentversion before use.

test plan II, which shall demonstrate a minimum acceptable MTBF of 750 hours. During each operating period, with each indicator properly connected, the following conditions shall exist:

a. Both axes of the attitude display shall be driven ±30° from 0° pitch and roll indications respectively. During the first 500 hours of operation on each indicator, the pitch axis shall receive a 10° per second input and the roll axis shall receive a 12° per second input. After 500 hours of operation on each indicator, the pitch axis shall receive a 1° per second input and the roll axis shall receive a 2° per second input during odd numbered operating periods and a 10° per second input in pitch and a 12° per second input in roll during even numbered operating periods. Within 10 minutes after the beginning of each operating period, the sphere shall follow the input signals without hunting, jumping, or interaction throughout the entire ±30° range or at the universal points.

b. The lighting circuits shall be energized with 5V ac or dc. During testing, the red and white lights shall be operated alternately every 100 hours.

c. The power failure flag shall disappear immediately after the beginning of each operating period.

* 4.6.42.1 After every 200 ±10 hours of operating time on each indicator and at the conclusion of reliability testing each of the six indicators shall pass all individual tests except 4.6.2.

4.6.43 Longevity. Testing of three of the six test samples specified in 4.6.42 shall be continued through a longevity test in accordance with MIL-STD-781, test plan XXVIII, for 2,500 hours on each indicator.

4.6.44 Production reliability. The three indicators selected for sampling plan B tests shall be subjected to a production reliability test as specified in 4.6.42.

4.6.45 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471, appendix B, test method 1, test plan A_1 plus B_2. Maintainability demonstration tests shall be performed on a minimum of three selected repairs. The tasks and the number selected shall be approved by the procuring activity.

4.7 Preservation, packaging, and packing. The inspection of preservation, packaging, and packing shall be in accordance the quality assurance provisions of MIL-STD-794 to insure compliance with section 5.

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:31Z
Check the source to verify that this is the current version before use.

* 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A (reference definition within MIL-STD-794). Indicators shall be packaged in accordance with Method IC-1 of MIL-P-116, one each in a unit container conforming to PPP-636, weather-resistant class. Each unit shall be cushioned with MIL-P-26514, class 2 polyurethane. Cushioning density, configuration, and thickness utilized shall insure that shock transmission to the physical item is dampened below the maximum permissible nonoperating g-limits of the item when completed packs are submitted to the rough handling tests of MIL-P-116.

5.1.2 Level C (reference definition within MIL-STD-794). Indicators shall be packaged identically to level A requirements specified in 5.1.1, except that method III of MIL-P-116 shall apply in lieu of method IC-1.

5.2 Packing

5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class. Insofar as practicable, exterior containers shall be of uniform shape and size and of minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in domestic-class exterior boxes conforming to PPP-B-636. The requirements specifie in 5.2.1 shall also apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection against physical and mechanical damage during direct domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking for shipment. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Indicator, Attitude, Remote Standby, 2 Inch __*__

* Applicable MS dash number to be added by contractor.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The remote attitude indicators covered by this specification are intended for use in fighter-type aircraft, trainers, and bombers as standby indicators.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. MS part number of indicator required (see 1.2)

c. When sampling plan B tests will not be conducted (see 4.5.2.2)

d. Levels of packaging and packing required (see section 5).

6.3 Definition

6.3.1 Hermetic sealing. Hermetic sealing is the process by which an item is totally enclosed within a suitable metal structure or case and sealed airtight by fusion of metallic or ceramic materials under heat or pressure and the fusion of metallic or ceramic materials into a metallic support. Elastomeric or resinous materials or combinations of these materials may be used as a cover glass seal provided the specified leak rate requirements are met under all tests and environmental conditions specified herein. These materials should be in accordance with MIL-S-7502 type B-2, or MIL-A-8623, type II or III. If these materials are used, the surfaces to be bonded must be cleaned in accordance with MIL-A-9067. A hermetic seal is not intended to include seals accomplished by gaskets.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention or suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ENFIF, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.

6.5 International standardization agreement. Certain provisions, figures 2 and 4, of this specification are the subject of international standardization agreements STANAG 3435 and ASCCAS 10/42. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing acitivity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.6 Marginal indicia. The outside margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activities:
Army -
Navy -
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 6610-0339

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:31Z
Check the source to verify that this is the current version before use.

4950/TZS
WRIGHT-PATTERSON AFB, O 45433

POSTAGE AND FEES PAID

UNITED STATES AIR FORCE
OFFICIAL BUSINESS

4950/TZS
Wright-Patterson AFB, O 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:31Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 66 WHICH MAY BE USED.

AFLC—WPAFB—OCT 67 2M

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:31Z
Check the source to verify that this is the current version before use.